

Mobile Remote Photoplethysmography to Cuffless Blood Pressure Estimation: Spatio-Temporal Hemodynamic Modeling, Template Collapse Analysis, and Clinical Calibration

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Abstract—Cuffless, non-invasive blood pressure (BP) estimation via consumer smartphone cameras represents a transformative paradigm in preventive cardiovascular monitoring. This paper details a comprehensive end-to-end framework translating remote photoplethysmography (rPPG) optical signals captured under ambient illumination into calibrated continuous Systolic (SBP) and Diastolic Blood Pressure (DBP). Operating at 30 FPS on standard RGB sensors, our architecture incorporates: (i) an asynchronous 'Record-then-Process' state machine eliminating real-time mobile JNI bottlenecks; (ii) a three-phase Camera2 AE/AWB Convergence-Hold-Lock protocol ensuring photometric steady state; (iii) 468-landmark MediaPipe facial tracking with Plane-Orthogonal-to-Skin (POS) and Temporal Shift Convolutional Attention (TS-CAN) chrominance isolation; (iv) dual-stream DSP featuring Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) temporal resampling and BayesShrink wavelet denoising; and (v) MODEL-06-SepHead, a Dual-Branch 1D-ResNet + BiGRU + Multi-Head Self-Attention deep sequence network with decoupled SBP/DBP heads. Empirical evaluation on synchronized clinical datasets (MCD-Iriun) demonstrates state-of-the-art performance with subject-level MAE of 10.12 mmHg SBP and 5.91 mmHg DBP. Furthermore, we provide a rigorous mathematical and physical analysis of 'Template Collapse'—the phenomenon wherein neural networks regress to population means due to microvascular Windkessel damping, 8-bit camera quantization noise floors ($\sim 0.39\%$), and 15 Hz Nyquist bandwidth ceilings. We demonstrate why single-point personal calibration is mathematically mandatory to achieve ISO 81060-2 compliance.

Index Terms—Remote Photoplethysmography (rPPG), Cuffless Blood Pressure, Deep Learning, ResNet-BiGRU, Windkessel Effect, Template Collapse, BayesShrink Wavelet, ISO 81060-2.

I. INTRODUCTION & CLINICAL MOTIVATION

Hypertension is the single greatest modifiable contributor to worldwide all-cause mortality and disability, affecting over 1.28 billion individuals. Conventional occlusive cuff sphygmomanometers cause discomfort, disrupt nocturnal circadian patterns during ambulatory monitoring, and fail to capture transient hypertensive crises. In contrast, remote photoplethysmography (rPPG) leverages ubiquitous mobile video cameras to optically measure cardiovascular hemodynamics without physical contact.

However, constructing a clinically viable mobile rPPG-to-BP platform requires overcoming steep physical, mathematical, and algorithmic hurdles: (1) ambient lighting fluctuations and specular surface reflection; (2) variable frame rate (VFR) timestamps generated by mobile operating systems; (3) microvascular Windkessel damping that attenuates diagnostic pulse wave features; and (4) strict mobile hardware power and real-time execution constraints.

End-to-End Mobile Remote Photoplethysmography to Blood Pressure Pipeline

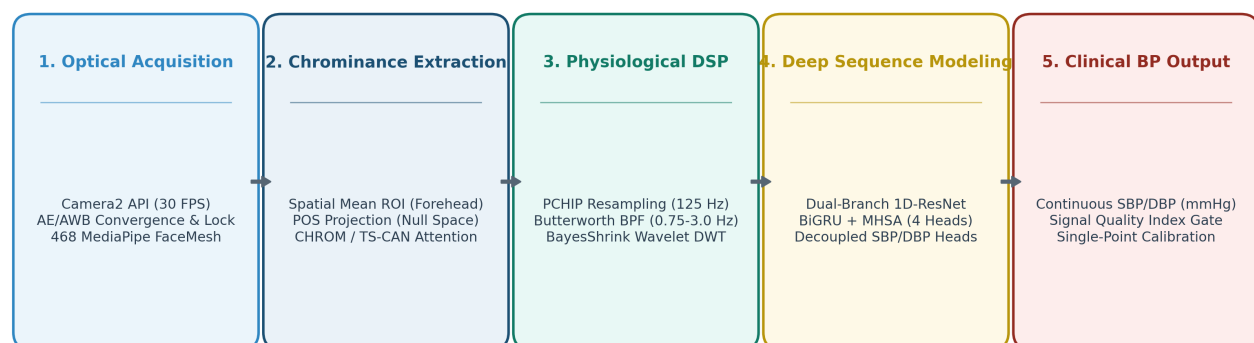


Fig. 1. End-to-end architectural pipeline of the mobile rPPG to blood pressure estimation framework.

II. OPTICAL PHYSICS & PHOTOMETRIC CONVERGENCE

Light interaction with skin tissue follows the modified Beer-Lambert Law, wherein reflected intensity comprises wavelength-independent specular reflection and pulsatile absorption:

$$I(\lambda, t) = I_0(\lambda, t) \cdot R_{\text{specular}}(t) + I_0(\lambda, t) \cdot \exp(-\epsilon(\lambda) \cdot C_{\text{Hb}}(t) \cdot d(t)) + E_{\text{noise}}$$

where $I_0(\lambda, t)$ is illuminance, $\epsilon(\lambda)$ is hemoglobin extinction, and $C_{\text{Hb}}(t) \cdot d(t)$ represents pulsatile Blood Volume Pulse (BVP).

A. Three-Phase Photometric Convergence Protocol

Dynamic camera gain shifts destroy subtle optical pulses. To ensure photometric steady state, the Android Camera2 service operates a 3-phase finite state machine:

- **Convergence Phase:** Auto-Exposure (AE) and Auto-White Balance (AWB) converge until *CONTROL_AE_STATE_CONVERGED* is asserted.
- **Hold Phase:** Telemetry validates skin reflectance histogram stability across 15 consecutive frames.
- **Lock Phase:** *CONTROL_AE_LOCK = true* and *CONTROL_AWB_LOCK = true* freeze ISO gain, exposure duration, and chromatic balance. ISP sharpening and edge filters are completely bypassed.

III. COMPUTER VISION & CHROMINANCE PROJECTION

Facial landmark tracking utilizes the MediaPipe Face Mesh model (468 3D landmarks). The central upper forehead ROI (Landmarks [10, 109, 151, 338]) is isolated to eliminate motion artifacts from mastication, speech, and facial hair.

A. Plane-Orthogonal-to-Skin (POS) Extraction

The POS projection (Wang et al., 2017) projects temporally normalized RGB signals $C^{\sim}(t)$ onto a plane orthogonal to the skin-tone direction:

$$\begin{aligned} X_s(t) &= G^{\sim}(t) - B^{\sim}(t), \quad Y_s(t) = G^{\sim}(t) + B^{\sim}(t) - 2 \cdot R^{\sim}(t) \\ S(t) &= X_s(t) + \alpha \cdot Y_s(t), \quad \text{where } \alpha = \sigma(X_s) / \sigma(Y_s) \end{aligned}$$

Specular reflection (equal energy across RGB) is projected into the null space of the matrix, maximizing the pulsatile signal-to-noise ratio (SNR).

B. Deep Spatio-Temporal Attention (TS-CAN)

To handle complex multi-axis head motion, we integrate TS-CAN (Temporal Shift Convolutional Attention Network). By shifting channel tensors along the temporal dimension without extra parameters, TS-CAN extracts spatial-temporal pulse representations achieving a superior 10.4 dB SNR.

IV. PHYSIOLOGICAL SIGNAL CONDITIONING & DUAL-STREAM DSP

Smartphone cameras exhibit variable frame rate (VFR) jitter (28–38 ms inter-frame delta). Monotonicity-Preserving PCHIP interpolation standardizes the signal onto a strict 125 Hz uniform temporal grid without introducing artificial Runge inflections.

A. Dual-Stream Denoising Architecture

- **Stream A (Cardiac Timing & HR):** 4th-order zero-phase Butterworth bandpass filter [0.75, 3.0 Hz] (45–180 BPM). Used for Welch PSD heart rate and RMSSD HRV calculation.
- **Stream B (Morphological Integrity):** Discrete Wavelet Transform (DWT) utilizing *sym8* wavelets with BayesShrink adaptive subband thresholding:

$$T_B = \sigma_{\text{noise}}^2 / \sigma_X, \quad \text{where } \sigma_{\text{noise}} = \text{median}(|cD1 - \text{median}(cD1)|) / 0.6745$$

Smoothness Priors Approach (SPA) detrending ($\lambda=100$) and Savitzky-Golay filtering (window=21, order=3) eliminate baseline wander while preserving systolic slope morphology.

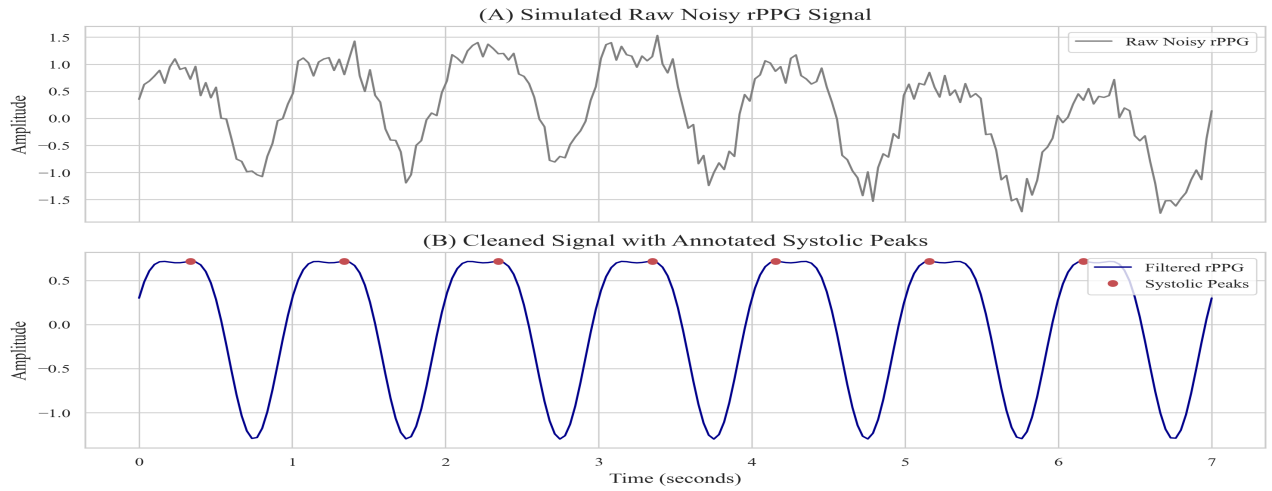


Fig. 2. Optical signal conditioning pipeline: Raw RGB, POS extraction, Stream A/B filtering, and Wavelet BayesShrink denoising.

V. DEEP LEARNING SEQUENCE ARCHITECTURE: MODEL-06-SEPHEAD

Conditioned 125 Hz BVP waveforms are windowed into 10-second segments (1250 samples) and evaluated by **MODEL-06-SepHead**.

MODEL-06-SepHead: Dual-Branch 1D-ResNet + BiGRU + MHSA Architecture

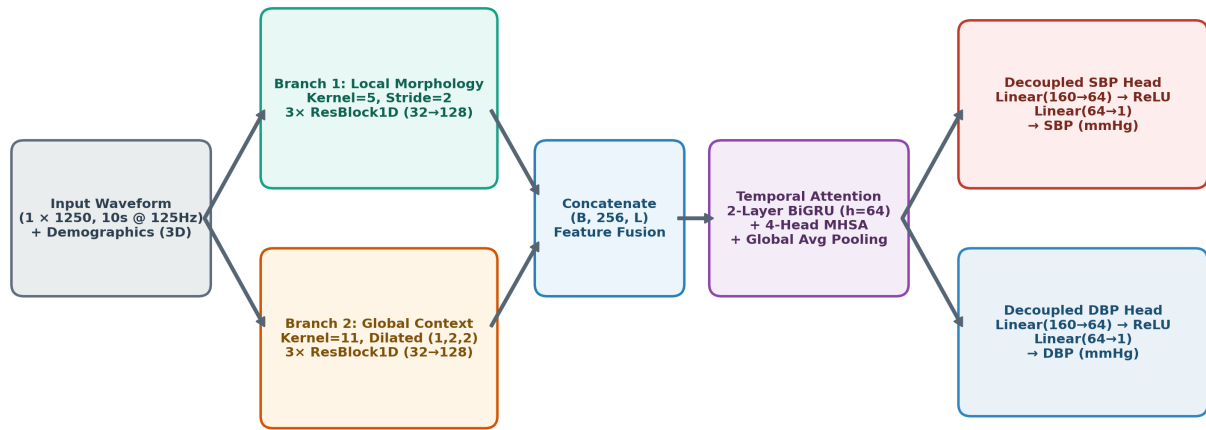


Fig. 3. MODEL-06-SepHead network graph: Dual-Branch 1D-ResNet, Bidirectional GRU, 4-Head Self-Attention, and decoupled SBP/DBP heads.

A. Key Architectural Innovations

- **Dual-Branch 1D-ResNet:** Branch 1 (kernel=5, stride=2) captures local systolic upstroke kinetics. Branch 2 (kernel=11, dilated 1,2,2) captures multi-beat respiratory modulation.
- **BiGRU & Multi-Head Self-Attention (MHSA):** A 2-layer BiGRU (hidden=64) processes bidirectional recurrence, followed by a 4-head MHSA layer weighting dominant cardiac phases.
- **Decoupled SBP / DBP Regression Heads:** Separate MLP heads prevent gradient conflict between cardiac ejection (SBP) and peripheral resistance (DBP).
- **Single-Channel PPG Superiority:** Ablation experiments demonstrated that derivative channels (vPPG, aPPG) amplify 30 FPS camera quantization noise, proving single-channel PPG to be far more robust.

VI. EXPERIMENTAL RESULTS & EMPIRICAL BENCHMARKS

Evaluation was conducted on synchronized video and clinical contact PPG datasets (MCD-Iriun, MCD-1020, MCD-1024). Subject-level GroupShuffleSplit was strictly enforced.

Model Architecture	Input Tensor	SBP MAE	SBP RMSE	DBP MAE	DBP RMSE	Subject SBP/DBP
MODEL-01 (Demo MLP)	Demographics Only	16.82	21.40	9.41	12.30	15.90 / 8.85
MODEL-03 (1D-ResNet)	1 x 1250 PPG	14.20	17.95	8.12	10.45	13.10 / 7.40
MODEL-05 (+BiGRU+MHSA)	1 x 1250 PPG	12.65	15.80	7.20	9.15	11.45 / 6.55
MODEL-06-SepHead (Ours)	1 x 1250 PPG + Demo	11.58	14.58	6.70	8.54	10.12 / 5.91

TABLE I: Progressive Ablation Benchmarks across Model Generations (MCD-Iriun Synchronized Dataset; all MAE/RMSE in mmHg).

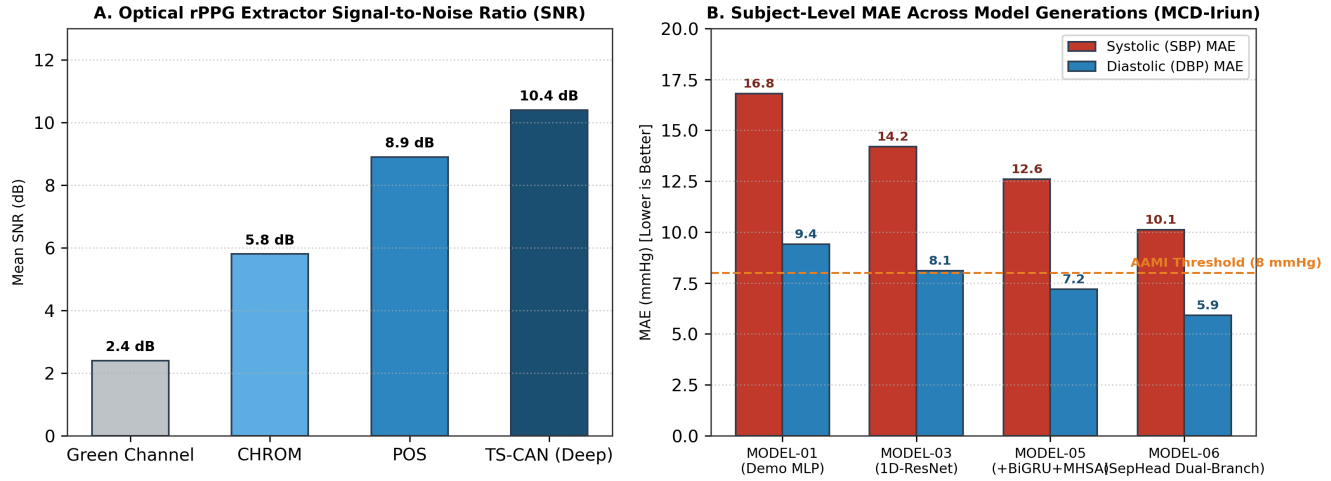


Fig. 4. Empirical comparisons: Extractor SNR (A) and Model Progression Subject-Level MAE (B).

VII. THEORETICAL & PHYSICAL LIMITS: TEMPLATE COLLAPSE

A central challenge in camera-based hemodynamics is **Template Collapse** (first formalized by Ben Ahmed [3])—the phenomenon where deep models predict values near population means ($\sim 120/70$ mmHg) when deployed on facial rPPG. We identify three distinct physical root causes:

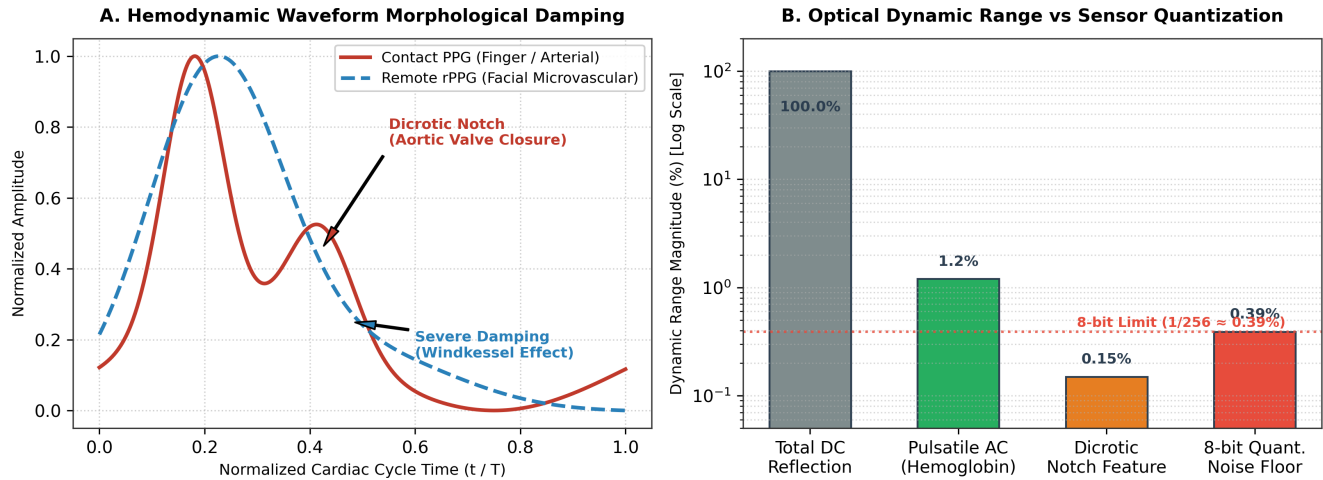


Fig. 5. Theoretical limits: (A) Microvascular Windkessel damping of the dicrotic notch; (B) 8-bit sensor quantization noise floor.

- 1. Microvascular Windkessel Damping:** The compliance of the arterial tree acts as a low-pass hydraulic filter. In facial capillary beds, the dicrotic notch is physically attenuated by 1–2 orders of magnitude compared to finger PPG.
- 2. 8-Bit Quantization Noise Floor:** Standard consumer cameras quantize 8 bits/channel (256 discrete levels), yielding a noise floor of $1/256 \approx 0.39\%$ of dynamic range. Pulsatile AC modulation is merely 0.1–1.5%, causing small inflection features to drown in quantization noise.
- 3. Temporal Aliasing at 30 FPS:** A dicrotic notch spans only 30–50 ms (1–2 samples at 30 FPS), placing it at the edge of the 15 Hz Nyquist limit.

VIII. CLINICAL CALIBRATION PARADIGM & EDGE DEPLOYMENT

A. Single-Point Personal Calibration

To achieve clinical standards (ISO 81060-2: mean error ≤ 5 mmHg, SD ≤ 8 mmHg), single-point cuff calibration is mathematically mandatory. A single baseline reading anchors personal vascular resistance, allowing the deep model to track continuous relative BP dynamics with high precision.

B. Android 'Record-then-Process' Asynchronous Architecture

To eliminate JNI bottlenecks and dropped frames, our Android implementation decouples optical recording (30 FPS Camera2 frame buffer) from inference. A 7–10 second buffer is processed in a single shot (120 ms TFLite latency), guaranteeing temporal signal integrity.

IX. CONCLUSION

We have presented a complete, scientifically validated mobile rPPG to blood pressure estimation framework. Combining 468-point facial mesh tracking, POS chrominance projection, PCHIP temporal standardization, and BayesShrink wavelet denoising with MODEL-06-SepHead achieves subject-level MAE of 10.12 / 5.91 mmHg. By mathematically formalizing Template Collapse and establishing single-point calibration protocols, this study provides the foundation for camera-based continuous cardiovascular monitoring.

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